**University of Dhaka**
Affiliated Faridpur Engineering College
Department of Computer Science and Engineering
**1st year semester 1st, 1st In course Examination -2024 (Session: 2023-24)**

**Time: 50 mints** **Marks: 20**

1. (a) Define the limit of a function using $(\delta-\varepsilon)$ .Using $(\delta-\varepsilon)$ definition prove that 1+2

$$\lim_{x\to 1}\frac{x^3-1}{x-1}=3$$

(b) Evaluate (any two) $(i)\lim_{x\to 0}(1+x)^{\frac{1}{x}}$ $(ii)\lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x}}$ $(iii)\lim_{x\to 0^+}x^{\frac{1}{x}}$ 1+1

(c) If $f(x)=|x-1|+|x+3|$ then show that $f(x)$ is continuous at $x=-3$ but not differentiable at $x=1$. 3

(d) A firm sells all the product it makes at TK12.00 per unit. The cost of making units $c(x)=20+0.6x+0.01x^2$. Find maximum profit. 2

2. (a) State Leibnitz's theorem. 1

(b) If $y=a\cos(\ln x)+b\sin(\ln x)$ , then show that $x^2y_{n+2}+(2n+1)xy_{n+1}+(n^2+1)y_n=0$ 3

(c) If $\varphi(x)=\begin{cases} x^2, when & x<1 \\ 2.5, when & x=1 \\ x^2+2, when & x>1 \end{cases}$ Dose $\lim_{x\to 1}\varphi(x)$ exist? 3

(d) Find the max and min values of $f(x)$ where $f(x)=\frac{4}{x}+\frac{36}{y}$ and $x+y=2$ 3